

# IEEE P2427 Analog Fault Weight Guidance

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# Fault Weights & IEEE Recommendations



# Why Do Analog Faults Need Weights?

- Silicon defects are traceable to the manufacturing process or to device aging and are related to device layout and geometry
- The probability of occurrence of defects can be estimated
- The probability of different types of defects in a circuit are not uniform
- Test coverage is improved if more likely faults can be detected
- IEEE P2427 recommends weighted test coverage
- CustomFault has a highly flexible capability to define fault weights

# Weighting Schemes Proposed by IEEE P2427

*Proposed in Annex of the Standard*

| Device    |                      | Scheme A                        | Scheme B                        | Scheme C                        |
|-----------|----------------------|---------------------------------|---------------------------------|---------------------------------|
| MOSFET    | Open G               | $P_{\text{inter\_open}} * S_1$  | 1                               | $P_{\text{inter\_open}} * S_1$  |
|           | Open D or S          | $P_{\text{inter\_open}} * S_1$  | 2                               | $P_{\text{inter\_open}} * S_1$  |
|           | Short DS             | $W * S_2 / (L * S_3)$           | $0.35 * W / L$                  | $W * S_2 / (L * S_3)$           |
|           | Short GD or GS       | $W * S_4$                       | W                               | $W * S_4$                       |
|           | Parametric           | $W * L * S_5$                   | Constant                        | $W * L * S_5$                   |
| Resistor  | Open                 | $R * S_6$                       | $0.35 * L / W$                  | $R * S_6$                       |
|           | Short                | $P_{\text{inter\_short}} * S_7$ | $0.35 * W / L$                  | $P_{\text{inter\_short}} * S_7$ |
| Bipolar   | Open B or E or C     | $P_{\text{inter\_open}} * S_1$  | $P_{\text{inter\_open}} * S_1$  | $P_{\text{inter\_open}} * S_1$  |
|           | Short BC or BE or CE | $P_{\text{inter\_short}} * S_7$ | $P_{\text{inter\_short}} * S_7$ | $P_{\text{inter\_short}} * S_7$ |
| Diode     | Open A or C          | $P_{\text{inter\_open}} * S_1$  | $P_{\text{inter\_open}} * S_1$  | $P_{\text{inter\_open}} * S_1$  |
|           | Short AC             | $P_{\text{inter\_short}} * S_7$ | $P_{\text{inter\_short}} * S_7$ | $P_{\text{inter\_short}} * S_7$ |
| Capacitor | Open                 | $P_{\text{inter\_open}} * S_1$  | 1                               | $P_{\text{inter\_open}} * S_1$  |
|           | Short                | $C * S_5$                       | $W * L * 1e12$                  | $C * S_5$                       |

| Variable                  | Scheme A | Scheme C    |
|---------------------------|----------|-------------|
| $S_1$                     | 1        | 25          |
| $S_2$                     | 3.5      | 2.857142857 |
| $S_3$                     | 10       | 0.5         |
| $S_4$                     | 1000000  | 1428571.429 |
| $S_5$                     | 1e15     | 1.65e13     |
| $S_6$                     | 0.00614  | 0.001       |
| $S_7$                     | 1        | 15          |
| $P_{\text{inter\_open}}$  | 1        | 0.1         |
| $P_{\text{inter\_short}}$ | 10       | 0.9         |

$P_{\text{inter\_open/short}}$  is probability of interconnect open or short

# Rationale Behind IEEE P2427 Weight Values

- Some faults are interconnect faults and their likelihood is directly related to the probability of a short or open in the interconnects. This is the case for:
  - MOS open (all terminals D, G and S)
  - BJT open (all terminals C, B and E)
  - Diode open (all terminals A and C)
  - Capacitor open (both terminals)
  - Resistor short between the 2 nodes
  - BJT or Diode short (BC, BE or CE)
- The weight of some faults is related to the device geometry
  - This is the case for MOS short
- The weight of some faults is related to the device value
  - This is the case for Resistor open and for Capacitor short
- When faults are related to the device geometry values, the  $\mu\text{m}$  values are normalized.
  - $S_4$  and  $S_5$  for example have large values

